VocalLens — Technical Report

Prototype: evidence-grounded contrastive speech-delivery analysis. **Event:** IIT Mandi Multimodal AI Hackathon — Track C (Contrastive Speech Analytics & Temporal Flaw Grounding). **Environment:** Windows 11, Python 3.11.9 (CPU only). **Report date:** 2026-10-14.

> Scope note: every number in this report is a *measured* value from this repository's current state. The evaluation was run on a **synthetic tonal pilot dataset**; it is feasibility evidence only and does **not** demonstrate generalization to real human speech. See §6.

1. Problem and approach

VocalLens compares a **participant** recording with a **reference** recording of the *same English text*, localizes measurable delivery differences in time, and explains each with the evidence behind it. Every reported deviation traces to a feature, a time region, a baseline comparison, a configured rule, a severity band, and an explicit evidence-quality assessment.

The product is a **comparison / coaching prototype**, not a judge of ability, intent, confidence, or emotion. A difference from the reference may be intentional or stylistic; the system reports measured deltas, not verdicts.

Design principles (enforced in code and tests)

- Forced alignment maps supplied words to audio; it does **not** verify transcript correctness.
- Reference and participant are aligned **independently** and compared on their own clocks, over **mapped phrase regions**, not identical absolute seconds.
- Missing/unreliable measurements are `null`/unavailable, **never zero**.
- Timestamps are never fabricated when alignment is unavailable.
- Scores are prototype rubric scores, not an objective measure of quality.

2. Architecture

A single pipeline contract is the one source of truth; the dashboard, evaluation script, and exporters all call it:

```
def analyze_pair(request: AnalysisRequest) -> AnalysisResult: ...
```

Stages (`src/vocallens/`):

Stage	Module	Responsibility
Validate / decode	`audio_io.py`	decode to 16 kHz mono analysis copy; record sample rate, channels, peak, clipping; reject corrupt/out-of-limit files; **no** loudness normalization before energy analysis
Transcript	`transcript.py`	normalize shared text; tokenize words
Alignment	`alignment.py`	independent forced alignment per recording; `whisperx` (optional) / `provided` / explicit `unavailable` / `partial` / `failed`
Region mapping	`region_mapping.py`	Levenshtein-based transcript↔audio mapping with a mismatch safeguard; phrase windows on each recording's own clock
Features	`features.py`	word/syllable rate, pauses, pitch variability (semitone IQR), within-recording relative energy (dBFS), MFCC diagnostics; validity + signal-quality flags
Detection	`detection.py`	pace / pause / pitch-variation / energy-dynamics candidates; normalized exceedance `R = D/T`; severity bands; same-family merge + cross-family `event_group_id` grouping
Quality	`quality.py`	evidence-quality index `Q = 0.35·A + 0.30·F + 0.20·S + 0.15·M` (ARCHITECTURE §8.7); unavailable when ungrounded
Explanations	`explanations.py`	plain-language, caveated explanation per detection
Scoring	`scoring.py`	prototype rubric from `configs/scoring.yaml`; piecewise-linear transform; ≥3-of-4 valid-dimension rule with weight renormalization
Export	`export.py`	versioned JSON + detections CSV
Synthesis	`synthesis.py`	deterministic generation of the synthetic pilot dataset

Schemas are typed (`schemas.py`); both YAML configs are validated on load (`config.py`) and reject invalid/unsupported values.

Scoring rubric (provisional, `configs/scoring.yaml`)

Four dimensions, piecewise-linear transform on aggregate normalized exceedance, anchored at `(0.0 → 100)`, `(1.0 → 70)`, `(3.0 → 0)`:

Dimension	Weight
pace	0.30
pause_management	0.30
vocal_variation	0.20
energy_control	0.20

Overall score requires **≥3 of 4** valid dimensions; weights are renormalized over the valid set. Below that threshold the overall score is returned as unavailable (not zero). These anchors are **not** tuned on held-out data.

3. Dataset

Synthetic, controlled pilot dataset (`data/README.md`). **It is not natural human speech.**

- **Recordings:** 36 (6 passages × 6 variants). 16 kHz mono PCM_16 WAV.
- **Passages:** 6 short original English sentences written for this repository.

- **Variants:** `reference`, `pace_fast`, `pace_slow`, `long_pause`, `monotone`, `energy_drop` (one reference + five controlled flawed variants).
- **Flaw families:** pace, pause, pitch_variation, energy_dynamics.
- **Splits (by passage, no leakage):** train = p01–p04 (24), val = p05 (6), test / held-out = p06 (6).
- **Ground truth:** each recording carries `word_timings` (known by construction, consumed via the `provided` alignment backend, labeled as such) and `flaw_intervals` (defined by the generating transform, not inferred after observing model output).
- **License / consent:** CC0-1.0; synthetic, no human recordings, no consent required. Provenance: generated deterministically by `vocallens.synthesis`.

Generation transforms (from `vocallens.synthesis`): `pace_fast` ×0.60 durations; `pace_slow` ×1.55; `long_pause` +1.1 s gap after the 2nd word; `monotone` constant F0; `energy_drop` ×0.30 amplitude on the 2nd half.

4. Alignment feasibility (honest status)

Only real-speech alignment *accuracy* remains BLOCKED / NOT VERIFIED; the previously-reported dependency conflict was re-investigated and retracted (`reports/alignment_feasibility.md`, revised 2026-10-13).

- **No dependency conflict (claim retracted).** `whisperx==3.8.6`, `streamlit==1.45.1`, `protobuf==6.33.6` (**< 7**), `numpy==1.26.4` (**< 2**), and `torch==2.8.0` were observed installed and importing together in one interpreter. WhisperX 3.8.6 declares plain `numpy` (unpinned); it does not force `numpy>=2.1` or `protobuf>=7`. WhisperX stays an optional extra (`requirements-align.txt`) only because it pulls a large stack (torch, etc.), not because of a version conflict.
- **Model asset loads.** `whisperx.load_align_model` downloads and loads the 360 MB English wav2vec2 alignment model on CPU (~112 s first run, then cached).
- **ffmpeg dependency removed.** The adapter previously failed at `whisperx.load_audio()` with `FileNotFoundError` (missing ffmpeg) and correctly degraded to `unavailable`. It now decodes via the project's soundfile/librosa decoder at the analysis rate, so ffmpeg is not required; the adapter then runs end to end (`whisperx.align()` executes and returns word-level timings).
- **Still not verified:** forced-alignment *accuracy* on real speech — no licensed/consented real English speech is available, and tonal fixtures have no phonetic content, so they are not a valid accuracy test. **No timestamps are fabricated** when alignment is unavailable.
- For the synthetic pilot the pipeline uses the `provided` ground-truth backend (labeled, not forced-alignment output).

5. Evaluation (measured)

Command: `python scripts/run_evaluation.py --split test` Source: `reports/evaluation/evaluation_test_latest.json` Generated: 2026-10-14T15:14:51Z · split `test` · **6 held-out cases** (passage p06). `is_synthetic_pilot = true`. (Detection metrics and per-case results are deterministic across runs; only the stage runtimes in §5.3 vary run to run.) Results below are **after** adding the utterance-level pitch-variation rule (ARCHITECTURE §8.6 "Pitch-variation intervals"); see §5.5 for the before/after.

> Report note (verbatim from the run): "*Feasibility pilot on synthetic tonal fixtures; NOT evidence of generalization to real speech. Thresholds were not tuned on this set.*"

5.1 Per-family detection metrics

Family	TP	FP	FN	Precision	Recall	F1
pace	2	1	0	0.667	1.000	0.800
pause	1	0	0	1.000	1.000	1.000
pitch_variation	1	0	0	1.000	1.000	1.000
energy_dynamics	0	0	1	—	0.000	—

5.2 Per-case detail

Case	Intended	Detected	#Det	Best IoU (intended)	Ref/Par coverage	Overall score	Status
p06_pace_fast	pace	pace	2	0.693	1.0 / 1.0	79	complete
p06_pace_slow	pace	pace	2	0.693	1.0 / 1.0	86	complete
p06_long_pause	pause	pace, pause	2	0.374	1.0 / 1.0	80	complete
p06_monotone	pitch_variation	pitch_variation	1	0.967	1.0 / 1.0	96	complete
p06_energy_drop	energy_dynamics	—	0	0.000	1.0 / 1.0	99	complete
p06_reference_self	none	—	0	n/a	1.0 / 1.0	100	complete

5.3 Runtime by stage (seconds, over 5 timed analyses)

Stage	Mean	Max
validate / decode	0.0040	0.0108
alignment	0.0003	0.0006
region mapping	0.0001	0.0001
features	0.6411	3.1168
detection	0.0022	0.0024
scoring	0.0001	0.0001
total	**0.7566**	**3.5383**

Feature extraction dominates runtime; alignment is near-free because the pilot uses the `provided` backend.

5.4 Reading the results (honest interpretation)

- **Pace and pause families fire correctly** on p06 (recall 1.0), with usable temporal localization (IoU 0.69 for pace; 0.37 for the inserted pause). The single pace false positive and the pace flag on `long_pause` reflect that an inserted pause also perturbs local word rate — association, grouped under a shared `event_group_id`, not a causal claim.

- **Pitch variation now fires** on `p06_monotone` (recall 1.0, precision 1.0, IoU 0.967) via the whole-utterance semitone-IQR rule added in §8.6. Monotone delivery reduces pitch range *across* the utterance (ref IQR 2.84 → participant 0.028 semitones); the per-phrase comparison alone diluted it below threshold. No thresholds were tuned on the held-out set — the rule compares the same semitone-IQR metric at utterance scale.
- **Energy-dynamics did not fire** on p06 (recall 0.0). Root cause is diagnostic, not a code defect: `z_rel` standardizes within-recording by design (to avoid gain/distance confounds), so the synthetic `energy_drop` — largely a *global* loudness reduction (median −22 → −29 dBFS) — produces only a small per-phrase relative contrast (R≈0.21 < 1.0), compounded by an inflated recording energy IQR (~14 dB from the synthetic envelopes). Catching it would need a different (absolute/trend) energy rule and non-synthetic data; deferred (see §7).
- `reference_self` **(reference vs. itself) scores 100 with zero detections** — a sanity check that the pipeline does not invent deviations where none exist, and the utterance pitch rule adds no false positive here.
- Overall scores remain a rubric transform, not a validated quality measure.

5.5 Before/after the utterance-level pitch rule

Family	Before (per-phrase pitch only)	After (+ utterance rule)
pitch_variation	TP 0, FN 1, recall 0.0, F1 —	TP 1, FN 0, **recall 1.0, F1 1.0**
p06_monotone case	0 detections, IoU 0.000, score 98	1 detection, **IoU 0.967**, score 96
pace / pause / energy	unchanged	unchanged (no new false positives)

The change is a structural detection fix (new interval for an existing metric), not threshold tuning on the test split.

6. Tests

`python -m pytest -q` → **80 passed** (Python 3.11.9). Coverage spans transcript normalization, region mapping and mismatch boundaries, config validation, schemas, features, detection (including the utterance-level pitch rule and its no-false-positive guard), quality, scoring, export, alignment fallback/unavailable paths and the ffmpeg-free WhisperX decode, the end-to-end pipeline (short, silent, noisy, clipped, mismatched, partial, and failed cases), and the FastAPI boundary (`tests/test_api.py`: health, samples, config, sample/upload analysis, evaluation, exports, and upload-validation/error paths via the fastapi `TestClient`).

7. Limitations

- **Synthetic tonal fixtures**, not natural speech: no formants, coarticulation, real prosody, room acoustics, or microphone variation. Results are feasibility evidence only and do not generalize to real speakers.
- **Real-speech forced-alignment accuracy is NOT VERIFIED** (no licensed/consented real audio). The dependency conflict previously cited was retracted: WhisperX installs alongside the dashboard, the model loads, and the adapter runs without ffmpeg — only a valid real-speech accuracy test is missing (§4).
- **Scores are provisional and unvalidated**; thresholds are not tuned on held-out data; no psychometric claims.
- **Energy-dynamics detection is under-sensitive** on this pilot (recall 0.0 on p06); the metric is intentionally invariant to global loudness, so a synthetic uniform-amplitude drop is a poor test of it (see §5.4).
- **Pilot size is tiny** (6 held-out cases, single synthetic voice per passage). No inter-annotator agreement is reported because labels are defined by the generating transform, not by multiple human annotators.

8. Next steps for a real evaluation

1. Replace synthetic fixtures with **consented, licensed human recordings** (≥2 speakers, split by passage) under `data/private/` (gitignored).
2. Install WhisperX from `requirements-align.txt` in a **separate alignment environment**; run the 3-clip feasibility gate on real speech and report alignment coverage and temporal error.
3. Add an absolute/trend-based energy rule (or better, non-synthetic data) so energy-dynamics is testable, and calibrate any thresholds on a **validation** split (never the held-out test set), then re-run the held-out evaluation.
4. Report per-family metrics, temporal IoU/overlap, alignment coverage, failure modes, and runtime on the real set; treat results as feasibility evidence until sample size supports stronger claims.

9. Reproducibility

```
python -m venv .venv && . .venv/Scripts/activate   # (macOS/Linux: source .venv/bin/activate)
pip install -r requirements.txt                    # core engine + Streamlit
python scripts/build_dataset_manifest.py           # build synthetic pilot
python scripts/validate_dataset.py                 # Gate A: dataset validation
python -m pytest -q                                # Gate B: 80 tests (engine + API)
python scripts/run_evaluation.py --split test      # held-out metrics

# Primary front end — React + FastAPI (two terminals):
pip install -r requirements-api.txt                # adds fastapi + uvicorn
uvicorn api.main:app --port 8000                   # terminal 1: API on :8000
cd frontend && npm install && npm run dev          # terminal 2: UI on :5173

# Fallback front end — Streamlit (no Node needed):
streamlit run app/streamlit_app.py                 # Gate C: dashboard
```

Each analysis JSON export carries software/config/model versions, config hashes, and per-stage timings for provenance. Canonical contracts live in `CLAUDE.md`, `ARCHITECTURE.md`, and `DESIGN.md`.